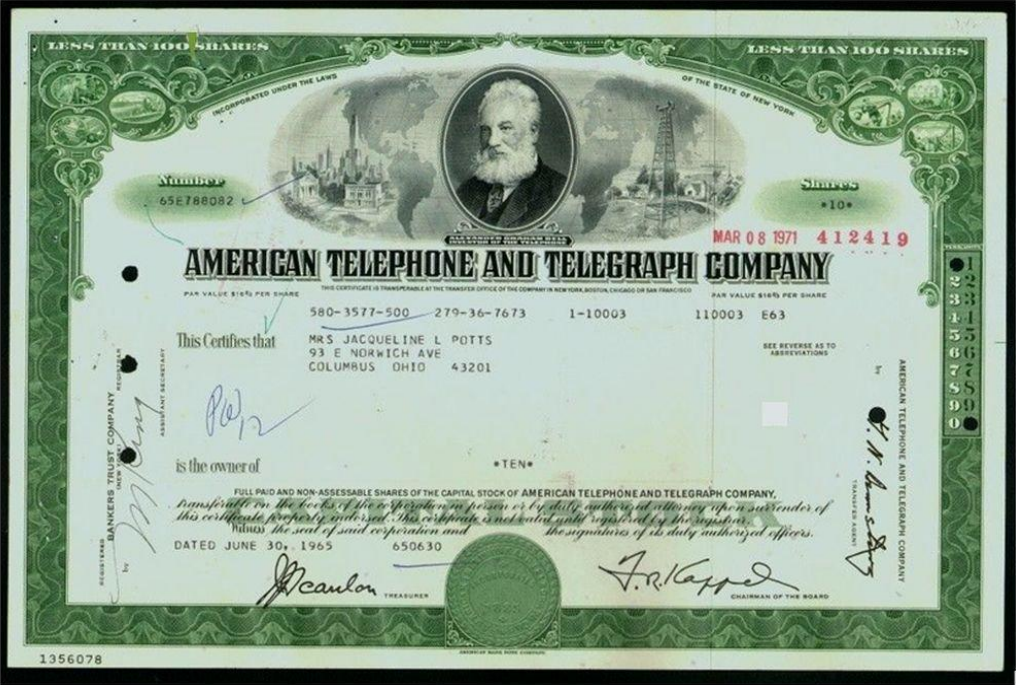


Dennis Ritchie & Kenneth Thompson

UNIX & C

Dennis Ritchie

Kenneth Thompson



AMA

Dennis Ritchie

Bell Labs R&D

Budget 70`s

\$ 500 million

Equivalent

\$ 1,5 billion

Today

Google is about

\$ 10 billion

Darpa budget

\$ 3 billion

Intel \$ 12 billion

AMA

Kenneth Thompson

Dennis Ritchie

Bell Labs

Nine Nobel Prizes

Four Turing Awards

Transistor

Laser

photovoltaic cell

UNIX

C Languageetc

AMA

Kenneth Thompson

Dennis Ritchie

Kenneth Thompson



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Dennis Ritchie

Kenneth Thompson



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Dennis Ritchie

Kenneth Thompson

DPD-7

Introductory price \$72,000

equivalent \$ 620,000 in 2021

UNICS OS

18 - bit

4 K

expandable to 64 K

AMA

Dennis Ritchie

Kenneth Thompson

Dennies Ritcher

Harvard University

Degrees physics and applied mathematics

start Phd but never finished

with support on B

programming language and

assembly creat C

AMA

Dennis Ritchie

Kenneth Thompson

never married

collaborated on M-209

used in Korean War

ciphertext - only attack

died one week after Steve Jobs

family in 2021 find Phd work

AMA

Dennis Ritchie

Kenneth Thompson

M-209



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Dennis Ritchie

Kenneth Thompson

```
Terminal
-rwxr-xr-x 1 sys      52850 Jun  8  1979 hptmunix
drwxrwxr-x 2 bin       320 Sep 22 05:33 lib
drwxrwxr-x 2 root     96 Sep 22 05:46 mdec
-rwxr-xr-x 1 root    50990 Jun  8  1979 rkunix
-rwxr-xr-x 1 root    51982 Jun  8  1979 rl2unix
-rwxr-xr-x 1 sys     51790 Jun  8  1979 rphtunix
-rwxr-xr-x 1 sys     51274 Jun  8  1979 rptmunix
drwxrwxrwx 2 root      48 Sep 22 05:50 tmp
drwxrwxr-x12 root     192 Sep 22 05:48 usr
# ls -l /usr
total 11
drwxrwxr-x 3 bin      128 Sep 22 05:45 dict
drwxrwxrwx 2 dmr       32 Sep 22 05:48 dmr
drwxrwxr-x 5 bin     416 Sep 22 05:46 games
drwxrwxr-x 3 sys     496 Sep 22 05:42 include
drwxrwxr-x10 bin     528 Sep 22 05:43 lib
drwxrwxr-x11 bin     176 Sep 22 05:45 man
drwxrwxr-x 3 bin     208 Sep 22 05:46 mdec
drwxrwxr-x 2 bin      80 Sep 22 05:46 pub
drwxrwxr-x 6 root     96 Sep 22 05:45 spool
drwxrwxr-x13 root    208 Sep 22 05:42 src
# ls -l /usr/dmr
total 0
#
```

AMA

Dennis Ritchie

Kenneth Thompson

Kenneth Thompson

bachelor and master degree in electrical engineering and computer
science

University California Berkeley

Multics project Thompson create a new language B to UNIX

In 1960 regular expression and have invented Thompson construction
algorithm

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Dennis Ritchie

Kenneth Thompson

Sabbatical from Bell labs helped install UNIX on PDP-11

Berkeley software distribution

16 bits

\$ 20.000 PDP-11 price

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Dennis Ritchie



Kenneth Thompson



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Dennis Ritchie

Kenneth Thompson

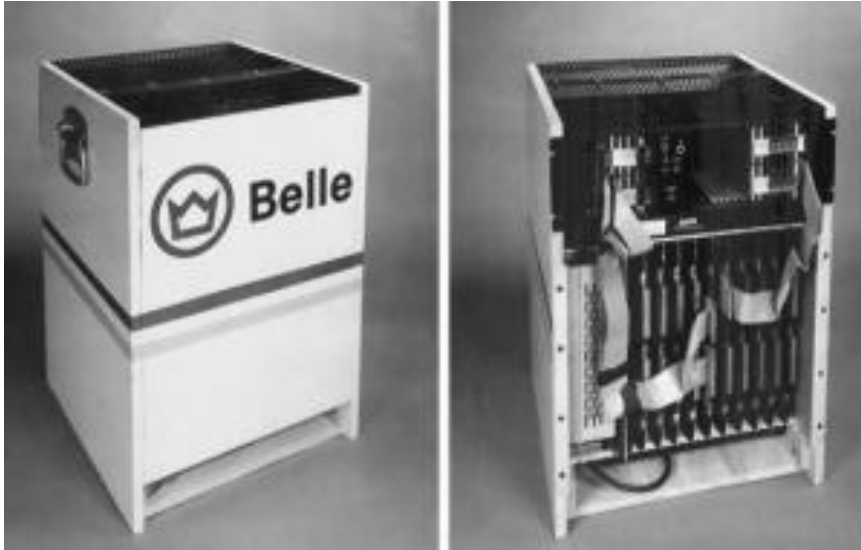
Bell chess machine

five time winner

ACM North American Computer

AMA

Dennis Ritchie



Kenneth Thompson

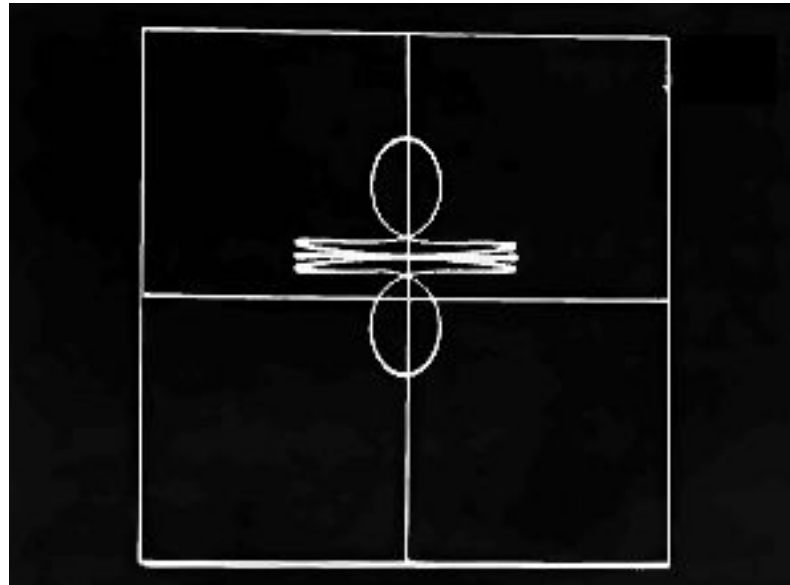


Dennis Ritchie

Kenneth Thompson

video game

star travel



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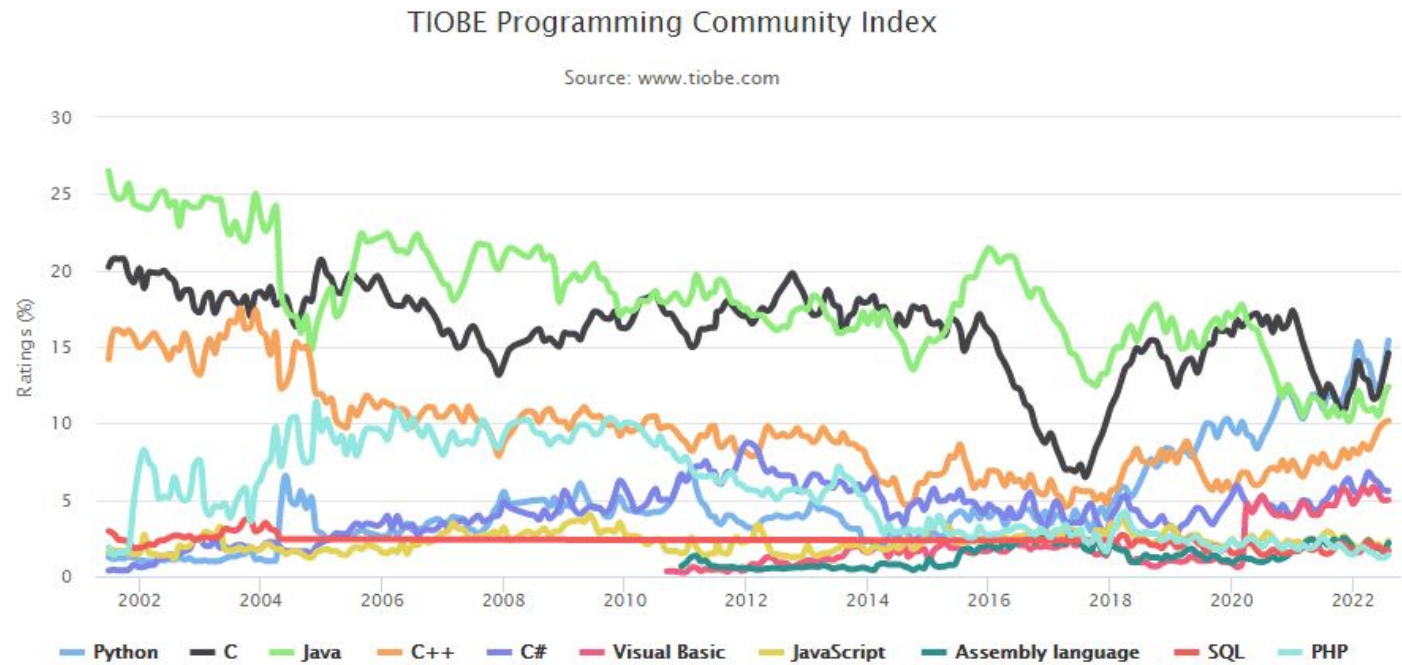
These are the ten most powerful supercomputers in the world, according to the TOP500 list (November 2022 update).

Supercomputer	Country	OS	Rmax	Rpeak
#1 Frontier	United States	HPE Cray OS	1,102 PFlops	1,685.65 PFlops
#2 Supercomputer Fugaku	Japan	Red Hat Enterprise Linux (RHEL)	442.01 PFlops	537.21 PFlops
#3 LUMI	Finland	HPE Cray OS	309.10 PFlops	428.70 PFlops
#4 Leonardo	Italy	Linux	174.70 PFlops	255.75 PFlops
#5 Summit	United States	RHEL 7.4	148.60 PFlops	200.79 PFlops
#6 Sierra	United States	RHEL	94.64 PFlops	125.71 PFlops
#7 Sunway TaihuLight	China	Sunway RaiseOS 2.0.5	93.01 PFlops	125.44 PFlops
#8 Perlmutter	United States	HPE Cray OS	70.87 PFlops	93.75 PFlops
#9 Selene	United States	Ubuntu 20.04.1 LTS	63.46 PFlops	79.22 PFlops
#10 Tianhe-2A	China	Kylin Linux	61.44 PFlops	100.68 PFlops

AMA

Dennis Ritchie

Kenneth Thompson



AMA

Dennis Ritchie

Kenneth Thompson

GO

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Dennis Ritchie

Kenneth Thompson

<https://www.cs.odu.edu/~tkennedy/cs300/development/Public/M01-HistoryOfComputers/index.html>

<https://blog.adafruit.com/2022/03/15/the-pdp-11-the-most-influential-minicomputer-of-all-time-arstechnica/>

<https://www.vitanuova.com/inferno/>

<https://www.oilshell.org/archive/Thompson-1968.pdf>

<https://www.bell-labs.com/usr/dmr/www/ken-games.html>

https://www.wikiwand.com/en/Dennis_Ritchie

<https://www.stackscale.com/blog/most-powerful-supercomputers-linux/>

AMA

Dennis Ritchie

Kenneth Thompson

AND

IoT

Android

Objective-C

Windows

Databases

Python C and UNIX are everywhere

END

AMA